

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS6

ODCS
BITOL @ LF

OpenAPI
LF

AsyncAPI
LF

GraphQL
LF

gRPC
CNCF

OData
OASIS

DATA PRODUCTS4

ODPS
BITOL @ LF

DPDS
ODM

ODPSpec
LF

DPROD
OMG

SCHEMA7

XML Schema
W3C

JSON Schema
IETF

SQL DDL
ISO/IEC

AVRO Schema
ASF

Protobuf
GOOGLE

LinkML
LINKML

Table Schema
FRICITIONLES S

SEMANTICS8

OSI
OSI INITIATIVE

RDF/OWL
W3C

DCAT
W3C

SKOS
W3C

SHACL
W3C

JSON-LD
W3C

schema.org
W3C

ShEx
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS8

CSV
IETF

JSON
IETF

XML
W3C

PARQUET
ASF

AVRO
ASF

ORC
ASF

Lance
LANCE

YAML
YAML.ORG

OPEN TABLE FORMATS4

Iceberg
ASF

Delta
LF

Hudi
ASF

Lance
LANCE

STORAGE SYSTEMS2

aws
S3
AWS

HDFS
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY4

JDBC
JCP

ODBC
ISO/IEC

ADBC
ASF

XMLA
MICROSOFT

INTERCONNECTION3

HTTP
IETF

FTP / SFTP
IETF

Delta Sharing
LF

MESSAGING5

Kafka
ASF

MQTT
OASIS

CloudEvents
CNCF

AMQP
OASIS

JMS
JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

DataFrame API
DATA-APIS

Apache Arrow
ASF

PROCESSING7

dbt
DBT LABS

Spark
ASF

Beam
ASF

Pandas
NUMFOCUS

Ibis
IBIS

SQL DML
ISO/IEC

XSLT
W3C

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS5

DuckLake
DUCKDB LABS

Iceberg Catalog
ASF

Unity Catalog
LF

Hive Metastore
ASF

Schema Registry
CONFLUENT

LINEAGE2

OpenLineage
LF

PROV
W3C

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY5

SQL
ISO/IEC

SPARQL
W3C

Substrait
LF

GQL
ISO/IEC

MDX
MICROSOFT

DATA QUALITY3

dbt
DBT LABS

Great Expectations
GX LABS

SodaCL
SODA

OBSERVABILITY2

Open Telemetry
CNCF

OORS
BITOL @ LF

POLICIES2

OPA
CNCF

ODRL
W3C

AI INTERFACES2

MCP
LF

A2A
LF